Lecture 26

Equivariant line bundles and Borel–Weil

Objectives. *Induced representations act on equivariant functions. The same functions can be read as sections of a vector bundle. In the Borel–Weil setting, holomorphic sections of an associated line bundle form a finite-dimensional representation. We compute these holomorphic sections completely for the projective line. The general Borel–Weil theorem is a cited extension.*

By the end of this lecture you will be able to:

- Compute all holomorphic sections of $\mathcal{O}(m)$ on $\mathbb{CP}^1$.
- Identify the resulting $\mathrm{SU}(2)$ representation and its central action.
- State Borel–Weil with a specified bundle character and dual convention.

26.1 From equivariant functions to sections

Let G be a Lie group, H a closed subgroup and $\sigma : H \to \mathrm{GL}(V)$ a continuous finite-dimensional complex representation. The quotient projection $G \to G/H$ is a submersion by Lecture 3. The local-section theorem [C: Lee, *Introduction to Smooth Manifolds*, second edition, Theorem 4.26] therefore gives a smooth local section near every point of G/H. We now attach a copy of V above each coset.

Definition 26.1. The *associated vector bundle* is

$$E = G \times_H V = (G \times V)/\!\sim, \qquad (gh, v) \sim (g, \sigma(h)v).$$

Its projection is $[g, v] \mapsto gH$. A *section* is a map $s : G/H \to E$ whose value at x lies in the fibre above x. If $\dim V = 1$, it is a *line bundle*.

Each fibre is a vector space: choose g over the coset and use $v \mapsto [g, v]$. Changing g changes this identification by an invertible linear map.

Choose smooth local sections $s_i : U_i \to G$ whose domains cover G/H. The map

$$U_i \times V \longrightarrow E|_{U_i}, \qquad (x, v) \longmapsto [s_i(x), v]$$

is a bijection, because every g over x has the form $s_i(x)h$ with $h \in H$. It is linear on each fibre. On $U_i \cap U_j$, write $s_j(x) = s_i(x)h_{ij}(x)$. The same bundle vector then has coordinates

$$v_i = \sigma(h_{ij})v_j.$$

The identity $h_{ij}h_{jk} = h_{ik}$ ensures consistency on triple overlaps.

A *smooth complex vector bundle* of rank k over a manifold M is a smooth manifold E with a smooth surjection onto M and a complex vector-space structure on each fibre E_x, the preimage of x. It is required to be locally a product. Each point of M has an open neighbourhood U and a diffeomorphism $E|_U \to U \times \mathbb{C}^k$ mapping each fibre E_x complex-linearly onto $\{x\} \times \mathbb{C}^k$. These diffeomorphisms are the *smooth local trivialisations*.

The bundle $E = G \times_H V$ has this structure. The local sections s_i are smooth, so the maps $h_{ij} = s_i^{-1}s_j$ are smooth into G. They take values in the embedded subgroup H, so they are smooth into H by the codomain-restriction result for embedded submanifolds [C: Lee, Corollary 5.30]. The representation σ is smooth by the automatic-smoothness theorem of Lecture 3. Hence the transition functions $\sigma(h_{ij})$ are smooth. Let $k = \dim V$, choose a basis of V, and regard $\mathbb{C}^k$ as $\mathbb{R}^{2k}$. Each fibre of E is then a real vector space of dimension $2k$, and the real matrices of $\sigma(h_{ij})$ are smooth maps into $\mathrm{GL}(2k, \mathbb{R})$. The vector bundle chart lemma [C: Lee, Lemma 10.6] applies to these real data. It gives E a unique topology and smooth structure for which the inverses of the maps above are smooth local trivialisations. These maps are complex-linear on fibres, so E is a smooth complex vector bundle.

This structure does not depend on the chosen family of local sections. For a second family (s'_j), the maps $s_i^{-1}s'_j$ are smooth into H by the same codomain-restriction argument. So the mixed transition functions $\sigma(s_i^{-1}s'_j)$ are smooth, and the chart lemma applies to the union of the two families. By its uniqueness clause, the union gives the same structure as each family alone.

A section is equivalently a function $f : G \to V$ with

$$f(gh) = \sigma(h)^{-1} f(g). \tag{26.1}$$

The correspondence is $s(gH) = [g, f(g)]$. It is well-defined because

$$[gh, f(gh)] = [g, \sigma(h) f(gh)] = [g, f(g)].$$

Conversely, a section determines a unique $f(g)$ in the coordinates supplied by g. Over U_i, the section is $x \mapsto [s_i(x), f(s_i(x))]$. The local trivialisation is a diffeomorphism, so s is smooth over U_i exactly when the coefficient $f \circ s_i$ is smooth. If f is smooth, then each $f \circ s_i$ is smooth. Conversely, suppose that each $f \circ s_i$ is smooth. Write $\pi : G \to G/H$ for the projection, and put

$$q(g) = s_i(\pi(g))^{-1} g \qquad (g \in \pi^{-1}(U_i)).$$

This map is smooth into G and takes values in H, so it is smooth into the embedded subgroup H by the same codomain-restriction result. Since $g = s_i(\pi(g))q(g)$, the covariance (26.1) gives

$$f(g) = \sigma\big(q(g)\big)^{-1} f\big(s_i(\pi(g))\big).$$

Hence f is smooth on each open set $\pi^{-1}(U_i)$, and therefore on G. The same argument with continuity in place of smoothness shows that s is continuous exactly when f is continuous.

We also need a holomorphic version of this construction. A function on an open subset of $\mathbb{C}^d$ is *holomorphic* if it is real C^∞ and its differential is $\mathbb{C}$-linear at every point. Polynomials are holomorphic, and so are quotients of polynomials where the denominator is nonzero. Composition of holomorphic maps is holomorphic by the chain rule.

Let Φ be a map between open subsets of $\mathbb{C}^d$ with holomorphic components and invertible differential at a point. The real inverse function theorem gives a smooth local inverse. Its differential is the inverse of a $\mathbb{C}$-linear map, hence $\mathbb{C}$-linear. So the local inverse is holomorphic. This is the *holomorphic inverse function theorem*. A *complex manifold* is a Hausdorff, second-countable topological space covered by charts. These charts are homeomorphisms of open sets onto open subsets of $\mathbb{C}^d$. The transition maps between them have holomorphic components. A map between complex manifolds is *holomorphic* if it is continuous and its expressions in charts have holomorphic components.

A *complex submanifold* of a complex manifold M is a subset N with the following property. Each point of N lies in a chart of M in which N is given by the vanishing of some coordinates. The remaining coordinates give charts of N. Hence a holomorphic map into M with values in N is also holomorphic as a map into N. We call this the *slice-chart property*.

A *complex Lie group* is a group with a complex manifold structure for which multiplication and inversion are holomorphic. The groups $K_{\mathbb{C}}$ of Lecture 14 are examples. A *closed complex subgroup* of a complex Lie group G is a closed subgroup that is also a complex submanifold.

Suppose that G is a complex Lie group, H is a closed complex subgroup and G/H is a complex manifold. Suppose also that the projection $\pi : G \to G/H$ is holomorphic, that σ is holomorphic and that the local sections s_i are holomorphic. Assume the U_i cover G/H.

If $g = s_i(x)h$, then $R_h \circ s_i$ is a holomorphic, hence smooth, local section through g, because $\pi \circ R_h = \pi$. By the local-section theorem of Lee, Theorem 4.26, π is a smooth submersion.

The topology of G/H is the quotient topology. Indeed, π is continuous, since it is holomorphic. Conversely, let $U \subset G/H$ be a set such that $\pi^{-1}(U)$ is open. Then $U \cap U_i = s_i^{-1}(\pi^{-1}(U))$ is open for each i, because $\pi \circ s_i = \mathrm{id}$. The sets U_i cover G/H, so U is open. By the uniqueness of the quotient structure in Lecture 3, the smooth structure underlying the complex manifold G/H is the quotient smooth structure.

In this case $h_{ij} = s_i^{-1} s_j$ is holomorphic as a map into G, because the group operations of G are holomorphic. By the slice-chart property, h_{ij} is holomorphic as a map into H. Then $\sigma(h_{ij})$ is holomorphic, as a composition of holomorphic maps.

Over the domain of a holomorphic chart φ of G/H inside U_i, the map $[s_i(x), v] \mapsto (\varphi(x), v)$ is a chart of E, with V identified with $\mathbb{C}^k$ by a basis. The change between two such charts has the form $(y, u) \mapsto (\psi(y), A(y)u)$. Here ψ is a transition map of G/H, and $A(y)$ is the matrix of $\sigma(h_{ij})$ at the point with coordinate y. The entries of A are holomorphic, so this change is holomorphic. Hence these charts make E a complex manifold. In this case E is a *holomorphic* vector bundle. The local trivialisations given by the sections s_i are its *holomorphic bundle charts*. For a second family (s'_j) of holomorphic local sections, the mixed changes of chart have the same form with $\sigma(s_i^{-1} s'_j)$, which is holomorphic by the slice-chart property. Hence both families give one holomorphic atlas, and the complex structure of E does not depend on the family. A section is *holomorphic* when each coefficient $f \circ s_i$ is holomorphic.

Left multiplication defines $a[g, v] = [ag, v]$. On sections its action is

$$(\Pi(a)s)(x) = a\, s(a^{-1}x).$$

In the function description this becomes $(\Pi(a)f)(g) = f(a^{-1}g)$. This is precisely the compact induction convention of Lecture 12.

26.2 The projective line and its two charts

Definition 26.2. The *complex projective line* $\mathbb{CP}^1$ is the set of one-dimensional complex subspaces of $\mathbb{C}^2$. A nonzero vector (v_0, v_1) determines the point $[v_0 : v_1]$, with $[cv_0 : cv_1] = [v_0 : v_1]$ for $c \neq 0$.

There are two charts. On $U_0 = \{v_0 \neq 0\}$ put $z = v_1/v_0$. On $U_1 = \{v_1 \neq 0\}$ put $w = v_0/v_1$. On the overlap, the transition between the charts is $w = 1/z$.

The charts give $\mathbb{CP}^1$ its topology. The points $[1 : 0]$ and $[0 : 1]$ have disjoint neighbourhoods $\{|z| < 1\}$ and $\{|w| < 1\}$. Every other pair lies in a common chart, so the space is Hausdorff. The two charts give a countable base. Each chart is a copy of $\mathbb{C}$, and the transition $w = 1/z$ is holomorphic on the overlap. Hence the two charts make $\mathbb{CP}^1$ a complex manifold, in the sense of the first section.

The group $\mathrm{SU}(2)$ acts on lines by matrix multiplication. In the z chart, the action of $g = \left(\begin{smallmatrix} a & -\bar{b} \\ b & \bar{a} \end{smallmatrix}\right)$ is

$$z \longmapsto \frac{b + \bar{a}z}{a - \bar{b}z}$$

where the denominator is nonzero. The other chart expressions are similar. These rational expressions show that the action is smooth. It acts transitively. Indeed, a unit vector $v = (a, b)$ is the first column of

$$g_v = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix} \in \mathrm{SU}(2).$$

The stabiliser of the line $\mathbb{C}(1, 0)$ is

$$T = \{t_\theta = \mathrm{diag}(e^{i\theta}, e^{-i\theta})\}.$$

By the homogeneous-space theorem of Lecture 3, $\mathbb{CP}^1 \cong \mathrm{SU}(2)/T$ as smooth manifolds. It is compact, as the image of the compact group.

The *tautological line bundle* has fibre the line itself:

$$\mathcal{O}(-1) = \{(\ell, v) : \ell \in \mathbb{CP}^1,\ v \in \ell\}.$$

Its dual is $\mathcal{O}(1)$. For $m \geq 0$, let $\mathcal{O}(m) = \mathcal{O}(1)^{\otimes m}$. Set $\mathcal{O}(0)$ equal to the trivial line bundle. For $m < 0$, use the $(-m)$th tensor power of $\mathcal{O}(-1)$.

The local frames of $\mathcal{O}(-1)$ are $e_0(z) = (1, z)$ and $e_1(w) = (w, 1)$. On the overlap, $e_1 = z^{-1}e_0$. The dual frames satisfy $e_1^* = ze_0^*$. For $m < 0$, $(e_i^*)^m$ means $e_i^{\otimes(-m)}$. Since $e_1^{\otimes(-m)} = z^m e_0^{\otimes(-m)}$, the same transition rule holds for $m < 0$. Hence a section of $\mathcal{O}(m)$ is a pair of functions f_0, f_1 with

$$f_0(z) = z^m f_1(1/z), \qquad f_1(w) = w^m f_0(1/w). \tag{26.2}$$

The frames $(e_0^*)^m$ and $(e_1^*)^m$ give fibrewise-linear bijections $\mathcal{O}(m)|_{U_i} \to U_i \times \mathbb{C}$. Their transition is multiplication by z^m, which is smooth and holomorphic on $U_0 \cap U_1$. The vector bundle chart lemma of Lee, Lemma 10.6, therefore makes $\mathcal{O}(m)$ a smooth line bundle with these trivialisations. We therefore call the section *holomorphic* when both functions are *entire*, that is, holomorphic on all of $\mathbb{C}$. The Borel–Weil discussion below shows that this definition agrees with the notion for associated bundles. We denote this space by $H^0(\mathbb{CP}^1, \mathcal{O}(m))$. The superscript 0 means ordinary global sections.

26.3 Computing the holomorphic sections

We import two results from complex analysis, the power-series theorem and Cauchy's estimate. If f is entire, then

$$f(z) = \sum_{k \geq 0} a_k z^k, \qquad |a_k| \leq R^{-k} \max_{|z|=R} |f(z)| \quad (R > 0).$$

Both are imported from [C: Stein and Shakarchi, *Complex Analysis*, 2003, Chapter 2, Section 4]. Theorem 4.4 and the paragraph after it state that an entire function has a power series about 0 that converges on all of $\mathbb{C}$, with $a_k = f^{(k)}(0)/k!$. The Cauchy inequalities, Corollary 4.3, give

$$|f^{(k)}(0)| \leq k! R^{-k} \max_{|z|=R} |f(z)|.$$

Dividing by $k!$ gives the estimate for a_k. These results apply to our entire functions, because a $\mathbb{C}$-linear differential makes a function complex differentiable.

Theorem 26.3. *For $m \geq 0$, the holomorphic sections of $O(m)$ are naturally the homogeneous polynomials of degree m on $\mathbb{C}^2$. In particular,*

$$\dim H^0(\mathbb{CP}^1, O(m)) = m + 1.$$

For $m < 0$, the space of holomorphic sections is zero.

Proof. Let (f_0, f_1) satisfy the transition rule (26.2). Since f_1 is bounded on $|w| \leq 1$, there is C with

$$|f_0(z)| \leq C|z|^m \qquad (|z| \geq 1).$$

If $f_0(z) = \sum a_k z^k$, Cauchy's estimate on a circle of radius $R \geq 1$ gives $|a_k| \leq CR^{m-k}$. Let $R \to \infty$. For $k > m$ this forces $a_k = 0$.

If $m \geq 0$, it follows that f_0 is a polynomial of degree at most m. Conversely every such polynomial works, because

$$f_0(z) = \sum_{k=0}^{m} a_k z^k \quad \Longrightarrow \quad f_1(w) = \sum_{k=0}^{m} a_k w^{m-k}$$

is entire and satisfies the transition. The corresponding homogeneous polynomial is

$$F(v_0, v_1) = \sum_{k=0}^{m} a_k v_0^{m-k} v_1^k.$$

Its chart coefficients are $F(1, z) = f_0(z)$ and $F(w, 1) = f_1(w)$. It obeys $F(cv) = c^m F(v)$ for $c \in \mathbb{C}$, so its restriction to each line is an element of that line's mth dual tensor power. Hence F is the coordinate-free description of the section.

If $m < 0$, every $k \geq 0$ satisfies $k > m$, so all a_k vanish. The transition then makes $f_1 = 0$ too. ∎

For $m = 2$, take $f_0(z) = a_0 + a_1 z + a_2 z^2$. Then $f_1(w) = a_0 w^2 + a_1 w + a_2$. Thus $F = a_0 v_0^2 + a_1 v_0 v_1 + a_2 v_1^2$.

26.4 The group action and its dual

A matrix $g \in \mathrm{SL}(2,\mathbb{C})$ sends a tautological fibre ℓ linearly to $g\ell$. The inverse transpose of this linear map sends the dual line ℓ^* to $(g\ell)^*$. Hence g acts on $O(1)$ and on all its tensor powers. Under the polynomial description, the section action is

$$(\Pi_m(g)F)(v) = F(g^{-1}v). \tag{26.3}$$

Indeed, at the line ℓ, the transformed functional evaluates a vector $v \in \ell$ by first pulling it back to $g^{-1}v \in g^{-1}\ell$. This gives the inverse in the formula.

The representation is $\mathrm{Sym}^m(\mathbb{C}^2)^*$, the dual of the symmetric tensor representation. For $\mathrm{SU}(2)$ this is isomorphic to V_m, by the prerequisite highest-weight theory. Here is also a direct weight check. On the basis $F_k = v_0^{m-k}v_1^k$,

$$\Pi_m(t_\theta)F_k = e^{i(2k-m)\theta}F_k, \qquad 0 \le k \le m.$$

The weights are $-m, -m+2, \dots, m$. They match V_m, with highest weight m and dimension $m+1$. Irreducibility is the symmetric-power result from Lie Algebras.

For a concrete matrix computation, put

$$g = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad g^{-1} = \begin{pmatrix} \bar{a} & \bar{b} \\ -b & a \end{pmatrix}.$$

For $m = 2$, in the ordered basis (v_0^2, v_0v_1, v_1^2), expansion gives

$$D^{(2)}(g) = \begin{pmatrix} \bar{a}^2 & -\bar{a}b & b^2 \\ 2\bar{a}\bar{b} & |a|^2 - |b|^2 & -2ab \\ \bar{b}^2 & a\bar{b} & a^2 \end{pmatrix}.$$

For example, the middle column comes from

$$(\bar{a}v_0 + \bar{b}v_1)(-bv_0 + av_1) = -\bar{a}b\,v_0^2 + (|a|^2 - |b|^2)v_0v_1 + a\bar{b}\,v_1^2.$$

The centre acts by $\Pi_m(-I) = (-1)^m I$. Thus the action on sections descends to $\mathrm{SO}(3)$ precisely for even m.

The same sections inside compact induction

For a homogeneous polynomial F of degree $m \ge 0$, a corresponding equivariant function $f_F : \mathrm{SU}(2) \to \mathbb{C}$ is given by $f_F(g) = F(g(1,0)^T)$. In the proof of Theorem 26.3, the section defined by F assigns to each line the restriction of F to that line. So $f_F(g)$ is the value of this restriction at the spanning vector $g(1,0)^T$. Since $gt_\theta(1,0)^T = e^{i\theta}g(1,0)^T$, homogeneity gives

$$f_F(gt_\theta) = e^{im\theta}f_F(g).$$

This is the covariance (26.1) for $H = T$ and $\sigma(t_\theta) = e^{-im\theta}$. Also

$$f_{\Pi_m(a)F}(g) = F(a^{-1}g(1,0)^T) = f_F(a^{-1}g),$$

so $F \mapsto f_F$ intertwines Π_m with the left-translation action $(\Pi(a)f)(g) = f(a^{-1}g)$ of the first section. A supplementary example in the solution of Exercise 26.5 identifies $O(m)$ with $\mathrm{SU}(2) \times_T \mathbb{C}_{-m}$, where T acts on $\mathbb{C}_{-m}$ by $e^{-im\theta}$.

26.5 The general Borel–Weil theorem

Let K be compact, connected, simply connected and semisimple. Its complexification $G = K_{\mathbb{C}}$ was introduced in Lecture 14. Sepanski, *Compact Lie Groups*, states Borel–Weil for a matrix model of G, so we first identify the two groups.

A faithful representation ρ of K exists [C: Bröcker and tom Dieck (BD), Section III.4, Theorem 4.1]. Averaging makes ρ unitary (Lecture 5). The entries of $r = \rho \oplus \overline{\rho} : K \to \mathrm{U}(N)$ generate $\mathscr{R}(K)$ [C: BD, Section III.4, Proposition 4.3].

Recall from Lecture 14 that each $\psi \in K_{\mathbb{C}} = \mathcal{X}(K)$ is a unital algebra homomorphism from $\mathscr{R}(K)$ to $\mathbb{C}$. The *coefficient map* $r_{\mathbb{C}}$ evaluates ψ on the matrix entries r_{ij} of r:

$$r_{\mathbb{C}}(\psi) = \big(\psi(r_{ij})\big)_{i,j=1}^{N}.$$

By [C: BD, Section III.8, Proposition 8.2 and the next paragraph], the map $r_{\mathbb{C}}$ sends $K_{\mathbb{C}}$ biholomorphically onto a closed subgroup and analytic submanifold of $\mathrm{GL}(N,\mathbb{C})$.

The resulting complex structure on $K_{\mathbb{C}}$ does not depend on r. Let r' be a second faithful unitary representation of K whose entries also generate $\mathscr{R}(K)$. Let $r'_{\mathbb{C}}$ be its coefficient map, defined by the same construction. Each entry of r' is a polynomial in the entries of r, and conversely. Every $\psi \in K_{\mathbb{C}}$ is a unital algebra homomorphism, so it respects these polynomial relations. Hence $r'_{\mathbb{C}} \circ r_{\mathbb{C}}^{-1}$ and its inverse are restrictions of polynomial maps. By the slice-chart property, they are holomorphic between the two analytic submanifolds. This common structure is the complex structure of Lecture 14.

The Lie algebra $\mathfrak{g}$ of $r_{\mathbb{C}}(K_{\mathbb{C}})$ is the complex span of that of $r(K)$ [C: BD, Proposition 8.4 and the preceding paragraph].

Since K is connected, so is $K_{\mathbb{C}}$ (Lecture 14). A connected Lie subgroup of $\mathrm{GL}(N,\mathbb{C})$ is determined by its Lie algebra [C: Kirillov Jr., *An Introduction to Lie Groups and Lie Algebras*, Theorem 3.40]. Hence $r_{\mathbb{C}}(K_{\mathbb{C}})$ is the complexification of $r(K)$ in Sepanski, Definition 7.48. We identify G with $r_{\mathbb{C}}(K_{\mathbb{C}})$ and K with $r(K)$, since $r_{\mathbb{C}}(\mathrm{ev}_g) = r(g)$.

Sepanski's construction applies to every faithful unitary r.

Fix a maximal torus T of K and positive roots. The *positive Borel subalgebra* $\mathfrak{b}$ is the complex Cartan subalgebra plus the positive root spaces. Let B be the connected Lie subgroup of G with this Lie algebra (Sepanski, Definition 7.48). By Sepanski, Lemma 7.49, B is closed. By Sepanski, Theorem 7.50, the inclusion induces a diffeomorphism from K/T onto the *flag manifold* G/B.

Since B is closed, it is an embedded Lie subgroup of G by the closed-subgroup theorem of Lecture 3. The inclusion of B with Sepanski's Lie subgroup structure is smooth into this embedded submanifold by Lee, Corollary 5.30. It is a bijective Lie group homomorphism, so it is a Lie group isomorphism by Lee, Corollary 7.6. Hence the embedded group B has Lie algebra $\mathfrak{b}$, and exp maps some neighbourhood O of 0 in $\mathfrak{b}$ diffeomorphically onto an open subset of B.

Choose a complex complement $\mathfrak{m}$ of $\mathfrak{b}$ in $\mathfrak{g}$. For $b \in B$, define $\Phi : \mathfrak{b} \times \mathfrak{m} \to G$ by

$$\Phi(X,Y) = b \exp X \exp Y.$$

The matrix exponential is holomorphic and maps $\mathfrak{g}$ into G, so Φ is holomorphic into G by the slice-chart property. Its differential at $(0,0)$ is the complex-linear isomorphism $(X,Y) \mapsto b(X+Y)$. By the holomorphic inverse function theorem, after shrinking O, there is a neighbourhood O' of 0 in $\mathfrak{m}$ such that Φ maps $O \times O'$ biholomorphically onto an open set $W' \subset G$. The set $\Phi(O \times \{0\}) = b \exp O$ is open in B. Because B has the subspace topology, it equals $B \cap W''$ for some open set W''. On $W = W' \cap W''$, the set B is exactly the slice $Y = 0$. Hence B is a closed complex subgroup.

The complex quotient theorem [C: Kirillov Jr., Theorem 2.11 and its proof] makes G/B a complex manifold, with holomorphic projection $\pi : G \to G/B$ and holomorphic local sections. Thus the first section applies to G, B and G/B.

Let λ be an integral weight. Since K is simply connected, $X^*(T) = P$ (Lecture 7). Let N_+ be the connected subgroup of G whose Lie algebra is the sum of the positive root spaces. By Sepanski, Lemma 7.49(d), each $b \in B$ is uniquely a product $t \exp(iH)n$ with $t \in T$, $H \in \mathfrak{t}$ and $n \in N_+$. These factors depend smoothly on b.

Our character of weight λ is $\exp H \mapsto e^{i\lambda(H)}$, so Sepanski's weight is our $i\lambda$ on $\mathfrak{t}$ (Sepanski, Definitions 6.23 and 6.26, Theorem 6.27 and formula (6.28)).

In our labels, Sepanski, Definition 7.52, defines $\chi_\lambda : B \to \mathbb{C}^\times$ by

$$\chi_\lambda\big(\exp(H')\exp(iH)n\big) = e^{i\lambda(H')-\lambda(H)} \qquad (H, H' \in \mathfrak{t},\ n \in N_+).$$

We import from Sepanski, Definition 7.52, the fact that this formula defines a homomorphism. The formula is well defined on $T = \exp \mathfrak{t}$ because $\lambda \in P = X^*(T)$. By the smooth dependence of the factors on b, the homomorphism χ_λ is smooth. It is trivial on N_+. Its differential at e is the complex-linear extension of $i\lambda$ on $\mathfrak{t}_{\mathbb{C}}$ and zero on the positive root spaces, so it is $\mathbb{C}$-linear. Left translations of B are holomorphic, and χ_λ is multiplicative. So its differential is $\mathbb{C}$-linear at every point, and χ_λ is holomorphic.

Define

$$L_\lambda = G \times_B \mathbb{C}_{-\lambda}, \qquad (gb, u) \sim (g, \chi_\lambda(b)^{-1}u).$$

Sections correspond to functions with $f(gb) = \chi_\lambda(b) f(g)$. Let $s_i : U_i \to G$ be holomorphic local sections of $\pi : G \to G/B$ whose domains cover G/B. By the first section, they and the holomorphic character χ_λ give holomorphic bundle charts of L_λ. A section is holomorphic in that sense exactly when its function $f : G \to \mathbb{C}$ is holomorphic. The proof is the argument of the first section, with holomorphic in place of smooth. In it, the map $q(g) = s_i(\pi(g))^{-1}g$ on $\pi^{-1}(U_i)$ is now holomorphic into B by the slice-chart property. The covariance writes $f(g)$ on $\pi^{-1}(U_i)$ as the product of $\chi_\lambda(q(g))$ and $f(s_i(\pi(g)))$, and both factors are holomorphic. Hence f is holomorphic on each set $\pi^{-1}(U_i)$, and therefore on G. Left translation acts as before. Write $H^0(G/B, L_\lambda)$ for the space of holomorphic sections of L_λ.

Theorem 26.4 (Borel–Weil; cited). *With the preceding positive Borel and bundle convention, if λ is dominant integral, then*

$$H^0(G/B, L_\lambda) \cong V_\lambda^*$$

as a K-representation. This is finite-dimensional and irreducible. If an integral λ is not dominant, then $H^0(G/B, L_\lambda) = 0$.

We import the theorem from Sepanski, Theorem 7.58, for the groups G and B above. Sepanski's Definition 7.54 calls a section holomorphic when its function on G is holomorphic.

By the equivalence above, his holomorphic sections are ours. Our L_λ is his line bundle of index $-\lambda$, and this change of index produces the dual. For $\mathrm{SU}(2)$, take r to be the standard representation. Its four entries generate the representative functions, and $r_{\mathbb{C}}(K_{\mathbb{C}}) = \mathrm{SL}(2,\mathbb{C})$ (Lecture 14). Thus $G = \mathrm{SL}(2,\mathbb{C})$, and B is the subgroup of upper triangular matrices in G. The character is

$$\chi_m(b) = c^m, \qquad b = \begin{pmatrix} c & \beta \\ 0 & c^{-1} \end{pmatrix} \in B.$$

It is trivial on the unipotent matrices, where $c = 1$. On the diagonal matrices it is $\mathrm{diag}(c, c^{-1}) \mapsto c^m$, the character of weight m. The fibre at $\mathbb{C}(1,0)$ transforms by c^{-m}.

The map $gB \mapsto [g(1,0)^T]$ identifies G/B with $\mathbb{CP}^1$. Indeed, G acts transitively on lines, since $\mathrm{SU}(2)$ already does, and B is the stabiliser of the line $\mathbb{C}(1,0)$.

The preliminary note in the solution of Exercise 26.5 compares the two section spaces. With Theorem 26.3, it follows that $F \mapsto f_F$, with $f_F(g) = F(g(1,0)^T)$ for $g \in G$, is an isomorphism $H^0(\mathbb{CP}^1, \mathcal{O}(m)) \cong H^0(G/B, L_m)$ of $\mathrm{SU}(2)$-representations for $m \geq 0$. For $m < 0$, both spaces are zero. Hence our explicit calculation is exactly the case $\mathrm{SU}(2)$ of Borel–Weil.

Self-checks

1. What is wrong with choosing an arbitrary entire f_0 as a section of $\mathcal{O}(1)$?

Answer. The coefficient $f_1(w) = w f_0(1/w)$ must also extend holomorphically to $w = 0$. This forces f_0 to have degree at most one.

2. Which representation occurs in $H^0(\mathbb{CP}^1, \mathcal{O}(3))$?

Answer. It is $\mathrm{Sym}^3(\mathbb{C}^2)^* \cong V_3$, of dimension four and spin $3/2$. The central element $-I$ acts by the scalar $(-1)^3 = -1$, so this representation does not descend to $\mathrm{SO}(3)$.

3. Does L_λ always produce highest weight λ in our general convention?

Answer. No. Its section space is V_λ^*, of highest weight $-w_0\lambda$. For $\mathrm{SU}(2)$ these weights coincide; for general groups they need not.

Summary

Equivariant functions correspond to sections of associated bundles. On $\mathbb{CP}^1$ and for $m \geq 0$, the transition rule for $\mathcal{O}(m)$ identifies holomorphic sections with homogeneous polynomials of degree m. Borel–Weil extends this construction to flag manifolds.